

Gibbs posterior distributions¹

direct, model-free inference on interest parameters

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- Quantities of interest exist in the real world, they're generally not just “model parameters.”
- e.g., quantiles, moments, optimal classifiers,...
- One *can* specify a model and express the quantity of interest in terms of model parameters, but this leads to
 - maybe lots of nuisance parameters
 - potential issues with marginalization²
 - risk of model misspecification bias
- I consider this to be an *indirect* approach.
- Can we avoid these pitfalls by taking a *direct* approach?

²M. arXiv:1607.05051

- Outside of a Bayesian context, making direct inference on quantities of interest is standard/typical.
- Is there a “Bayesian version” of this direct approach?
- Yes, *Gibbs posterior distributions*
 - quantity of interest defined by minimizing risk
 - work with empirical risk instead of likelihood
- What kind of properties do Gibbs posteriors have?
 - concentration rates
 - uncertainty quantification

- Gibbs posterior primer
- Minimum clinically important difference
 - population version
 - personalized version
- Multivariate quantiles
 - Bernstein–von Mises
 - calibrating credible regions
- Concluding remarks

- Quantity of interest is defined as the minimizer of a risk function, that is, $\theta^* = \theta(P) = \arg \min R(\theta)$.
 - risk function is given, e.g., median = $\arg \min E|Y - \theta|$,
 - or I can cook up a suitable risk function myself.
- Apply Bayes's formula but with the negative log-likelihood replaced by a scaled empirical risk, i.e.,

$$\Pi_n(d\theta) \propto e^{-\omega n R_n(\theta)} \Pi(d\theta).$$

- Direct attack on the quantity of interest:
 - no nuisance parameters or marginalization
 - only needs a prior for the relevant quantity
 - no model, can't be misspecified
- Choice of learning rate ω is crucial — data-driven?

Minimum clinically important difference

- Data (X_i, Y_i) , $i = 1, \dots, n$, iid:³
 - $X_i \in \mathbb{R}$ is a diagnostic measure on patient i ;
 - $Y_i \in \{\pm 1\}$, depending on whether patient i found the treatment to be effective/ineffective.
- Quantity of interest is the MCID:

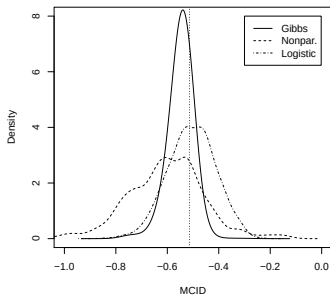
$$\theta^* = \theta(P) = \arg \min_{\theta} \underbrace{P\{Y \neq \text{sign}(X - \theta)\}}_{R(\theta)},$$

where P is the joint distribution of (X, Y) .

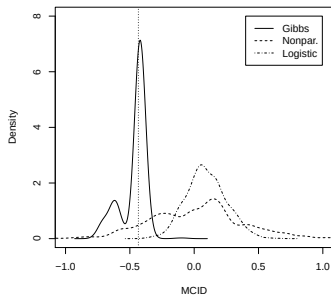
- MCID is the X -scale cutoff at which patients will tend to view the treatment as effective.
- Statistical versus practical/clinical significance...

³Syring and M., arXiv:1501.01840

- Lots of Bayesian things one could try, but...
- Gibbs is easy: $R_n(\theta) = n^{-1} \sum_{i=1}^n 1\{Y_i \neq \text{sign}(X_i - \theta)\}$.
- Compare Gibbs ($\omega = 1$) to two Bayes posteriors
 - logistic regression
 - binary regression with nonparametric link



(a) "Good case"



(b) "Bad case"

- MCID is also characterized as a solution to $\eta(x) = \frac{1}{2}$, where

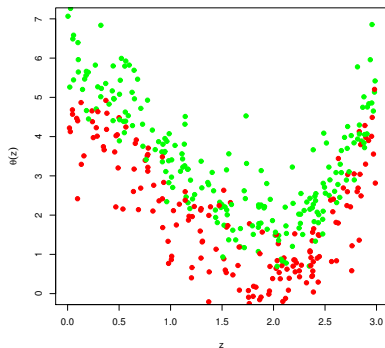
$$\eta(x) = P(Y = 1 \mid X = x).$$

Theorem (Syring and M., arXiv:1501.01840)

Assume, roughly, that $|\eta(x) - \frac{1}{2}| \approx c|x - \theta^*|^\gamma$. For any reasonable prior, if the learning rate ω is sufficiently small, then the Gibbs posterior concentration rate is roughly $n^{-1/(1+2\gamma)}$.

- Specific cases:
 - if η is differentiable, then $\gamma = 1$ and rate is $n^{-1/3}$
 - if η has a jump at θ^* , then $\gamma = 0$ and rate is n^{-1} .

- *Personalized medicine* is a trend, no “one size fits all.”
- Naturally, there's interest in a *personalized MCID*.
- Data (X_i, Y_i, Z_i) looks like this:



- MCID is now a function $\theta(z)$ of the covariate...

- Construction of the Gibbs posterior for θ is basically the same as before, i.e., risk counts missclassifications.
- Key difference is that θ is a function — nonparametric!
- Series prior for θ , e.g., using using B-splines.
- Two factors determine the rate:
 - γ : smoothness of $x \mapsto \eta_z(x)$ for $x \approx \theta^*(z)$
 - α : smoothness of $z \mapsto \theta^*(z)$

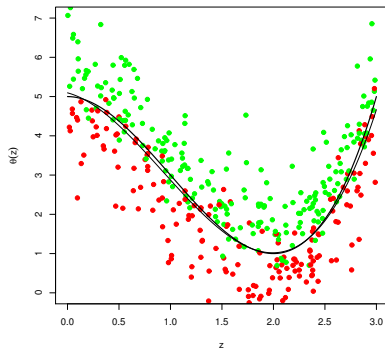
Theorem (Syring and M., almost finished)

For a suitable prior, if ω is sufficiently small, then, roughly,

$$\Pi_n(\{\theta : \|\theta - \theta^*\|_{L_1(P)} \gtrsim n^{-r(\alpha, \gamma)}\}) \rightarrow 0,$$

where $r(\alpha, \gamma) = \frac{1}{1+2\gamma} \wedge \frac{\alpha}{\alpha+1}$.

- Same data points, with two overlays:
 - true MCID curve θ^* (heavy)
 - Gibbs posterior mean curve (thin)



Multivariate quantiles

- In dimension ≥ 2 , the definition of “quantile” isn’t clear.
- One definition of a *multivariate median* is

$$\theta^* = \arg \min_{\theta} \mathbb{E}\{\|X - \theta\| - \|X\|\}.$$

- Bayesian inference is challenging because there’s no model, but Gibbs is straightforward:

$$\Pi_n(d\theta) \propto \exp\left\{-\omega \sum_{i=1}^n \|X_i - \theta\|\right\} \Pi(d\theta).$$

Theorem (Bhattacharya and M., almost finished)

Under mild conditions on P , if the learning rate ω is sufficiently small, then the Gibbs posterior has

- a $n^{-1/2}$ concentration rate and
- a Bernstein–von Mises/approximate Gaussian property.

- However, the covariance matrix in the BvM might not agree with that of the sampling dist for the M-estimator.
- Bias from cov mismatch affects uncertainty quantification.
- Highlights a key difference between Bayes and Gibbs:
 - Bayes has no remedy for this bias, short of starting over
 - with Gibbs, we can at least partially correct for this mismatch by adjusting the learning rate.
- *Question:* How to adjust the learning rate?

- Recall the Gibbs posterior:

$$\Pi_n(d\theta) \propto e^{-\omega n R_n(\theta)} \Pi(d\theta).$$

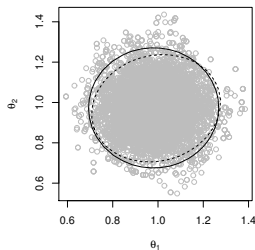
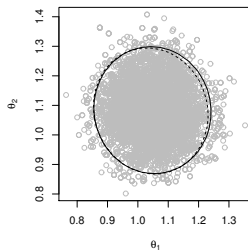
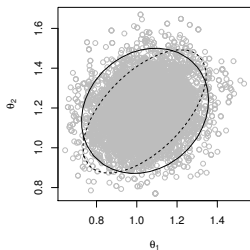
- Since R_n is minimized near $\theta^* = \arg \min R(\theta)$, Π_n is roughly centered there, so ω controls the spread around θ^* .
- ω too big makes credible sets too small and vice versa.
- Goal: valid Gibbs posterior uncertainty quantification.
- Can we tune ω so that Π_n credible regions are calibrated, e.g., 95% credible regions are 95% confidence regions?

General Posterior Calibration (GPC) Algorithm:

- 0 Initialize ω_0 and set $t = 0$
- 1 Sample new data sets from $\mathbb{P}_n$ to approximate the coverage prob of Gibbs credible region $C_{\omega_t, \alpha}$:
 - do Gibbs posterior computation for each new data set
 - check coverage using $\theta(\mathbb{P}_n)$
- 2 If coverage prob is close to $1 - \alpha$, then stop; otherwise, update $\omega_t \rightarrow \omega_{t+1}$ via stochastic approximation, increment t , and go back to Step 1.

(Syring and M., arXiv:1509.00922)

- A few cases with the multivariate median
 - Gibbs posterior credible region (solid)
 - M-estimator confidence region (dashed)
- Whole computation takes just a few seconds on my laptop.



- Two-dim quantile regression example⁴
- Several methods:
 - BEL.s Bayesian empirical likelihood
 - Normal asymptotic normality
 - GPC Gibbs with GPC
- GPC gives exact coverage and shorter length!

n		Coverage Prob $\times 100$			Avg Length $\times 100$		
		BEL.s	Normal	GPC	BEL.s	Normal	GPC
100	θ_0	97	95	95	106	100	91
	θ_1	98	98	95	58	55	47
400	θ_0	95	95	95	50	50	46
	θ_1	97	97	95	26	25	23
1600	θ_0	96	96	95	25	25	23
	θ_1	96	96	95	13	12	11

⁴Syring and M., arXiv:1509.00922

Concluding remarks

- Gibbs approach makes direct posterior inference possible:
 - eliminates misspecification bias
 - no nuisance parameters
 - no marginalization
- Good concentration rate properties.
- Can tune learning rate for valid uncertainty quantification
- Not just for cases where R is given, we can also reverse engineer a suitable R :
 - image segmentation (Syring and M., arXiv:1606.08400)
 - area under ROC curve (Wang and M., arXiv:1906.08296)
- More Gibbs stuff to do:
 - proof of the conjecture in bullet point #3 above
 - applications
 - computation

Thank you!

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